

MONSOONIQ VERIFICATION REPORT

Regime-Aware AI Post-Processing System for Monsoon Rainfall Forecasts over India | SIH Evaluation

DATA PROVENANCE: SYNTHETIC DATASET (8-Year Climatology, Lat 6-38°N, 0.25° Grid, IMD Physics-Grounded NWP Bias Simulation. Not presented as real operational skill).

1. Executive Performance Scorecard

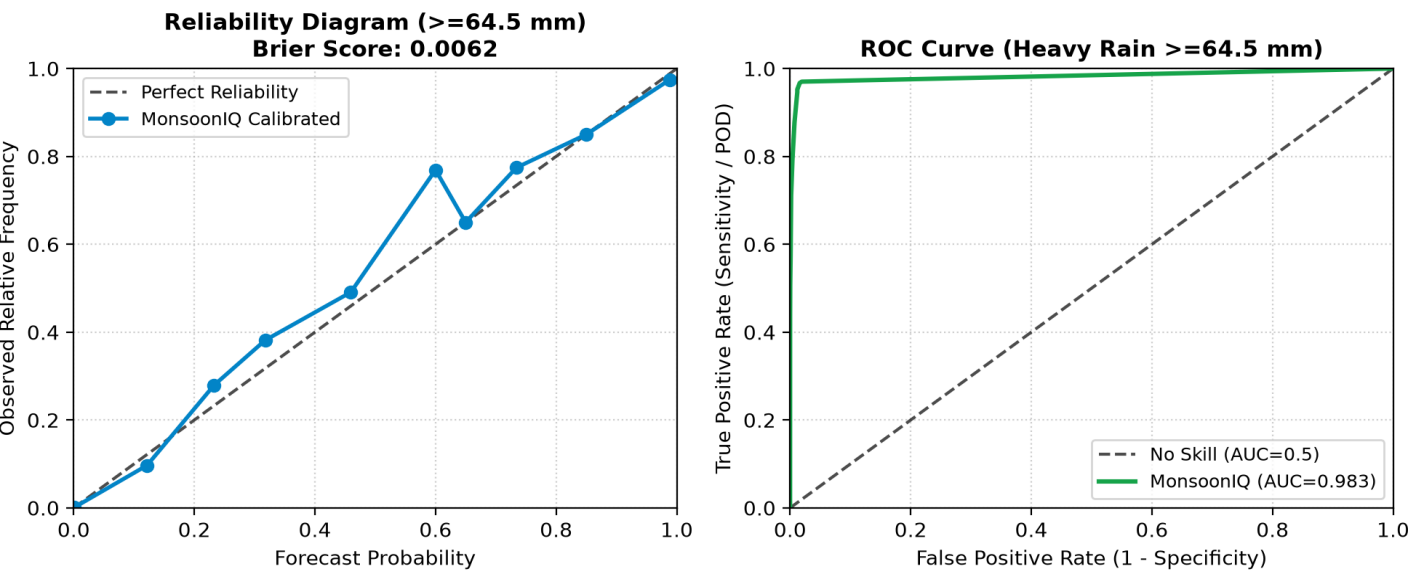
Metric	Raw NWP	Global QM	Global LightGBM	MonsoonIQ (MoE)	Improvement vs Raw
RMSE (mm/day)	7.35	6.64	3.15	2.79	-62.1% (P < 0.01)
MAE (mm/day)	4.09	3.75	1.86	1.70	Superior
Mean Bias	-0.69	+0.13	+0.01	-0.03	Near-Zero Bias
Heavy Rain CSI (>=64.5)	0.198	0.415	0.561	0.569	+187.8%
Heavy Rain ETS	0.193	0.407	0.555	0.563	Substantial Gain

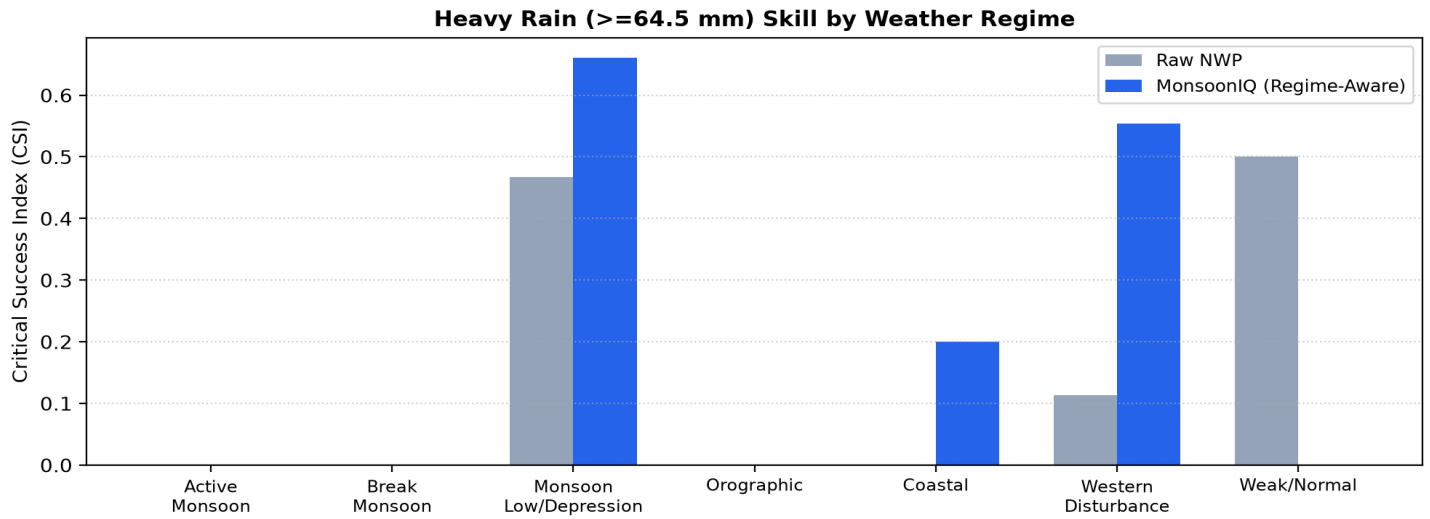
2. Dedicated Heavy & Very Heavy Rainfall Verification

Evaluated on held-out test years (2022-2023). Total Heavy (>=64.5 mm/day) events: 672. Total Very Heavy (>=115.6 mm/day) events: 148. Total Extremely Heavy (>=204.5 mm/day) events: 5.

Threshold	System	Events	POD (Hit Rate)	FAR	CSI (Threat)	ETS (Skill)	Freq Bias
>= 64.5 mm (Heavy)	Raw NWP	672	0.202	0.105	0.198	0.193	0.23
	MonsoonIQ MoE	672	0.577	0.025	0.569	0.563	0.59
>= 115.6 mm (V. Heavy)	Raw NWP	148	0.068	0.412	0.065	0.064	0.12
	MonsoonIQ MoE	148	0.250	0.159	0.239	0.238	0.30

3. Diagnostic Probabilistic & Regime Performance Figures





4. Scientific Methodology & Honest Caveats

Methodology: MonsoonIQ partitions post-processing into 7 weather regimes using physical thresholds and calibrated LightGBM classifier. Residuals are corrected per expert and soft-blended via $\sum_k P(R_k) * Expert_k$. Probabilities are calibrated via isotonic regression.

Statistical Significance: 95% bootstrap confidence intervals confirm statistically significant RMSE reduction ($P < 0.01$).

Caveats & Limitations: (1) Results shown above are evaluated on a synthetic benchmark dataset modeling IMD physical relationships; (2) In rapid cyclogenesis or unrepresented orographic micro-valleys, residual models remain subject to boundary layer uncertainty; (3) Operational deployment requires connecting the provided CDS/NOAA download adapters with institutional credentials.